

Instagram Research Agent README

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1 Project Overview

Brief description of the Instagram Research Agent.

Topics:

- Purpose of the project
- What the workflow does
- End-to-end automation overview

2 Features

2.1 Phase 1: My Reels Analysis

- Fetch Instagram Reels
- Sort top-performing reels
- Download videos
- Whisper transcription
- Hook analysis using GPT
- Google Sheets logging

2.2 Phase 2: Competitor Research

- Discover competitors
- Extract top 3 reels
- Analyze content
- Store results

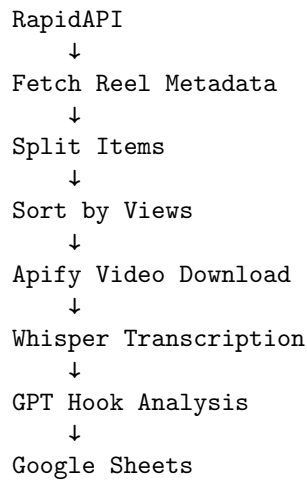
2.3 Phase 3: Niche Outliers

- Claude Sonnet agent
- Viral outlier detection
- Content pattern analysis

3 Tech Stack

- n8n
- RapidAPI Instagram Scraper
- Apify Instagram Scraper
- OpenAI Whisper
- GPT-4.1 Mini
- Claude Sonnet
- Google Sheets

4 Workflow Architecture



5 Prerequisites

5.1 n8n Setup

Explain:

- Install n8n or use cloud
- Import workflow JSON

5.2 RapidAPI Setup

- Create account
- Get API Key
- Configure headers

5.3 Apify Setup

- Create account
- Get API Token

5.4 OpenAI Setup

- Add API key
- Whisper configuration
- GPT configuration

5.5 Google Sheets Setup

- Create spreadsheet
- Add tabs:
 - My Reels
 - Competitor List
 - Niche Outliers

6 How to Set Up

6.1 Step 1: Import Workflow

Import the provided n8n workflow JSON.

6.2 Step 2: Configure Credentials

Set up:

- RapidAPI credentials
- Apify API token
- OpenAI API key
- Google Sheets authentication

6.3 Step 3: Configure Username

Replace the username field with your Instagram username.

7 How to Run

7.1 Manual Run

1. Open n8n
2. Execute workflow
3. Wait for completion
4. Check Google Sheets output

7.2 Automated Run

Workflow runs:

Every Sunday at 5:00 AM

8 Output Format

Describe:

- Reel URL
- Views
- Likes
- Comments
- Transcript
- Hook Analysis
- Thumbnail

9 Troubleshooting

Common issues:

- Missing video URL
- Whisper transcription failure
- Google Sheets authentication errors
- API rate limits

10 Future Improvements

- Better competitor discovery
- Advanced AI analysis
- Dashboard integration
- Automated reporting

11 References

- <https://n8n.io/>
- <https://rapidapi.com/>
- <https://apify.com/>
- <https://platform.openai.com/>